Lesson for the course "Human-Computer Interaction"

Visual perception (Part 2)

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Our color vision is limited

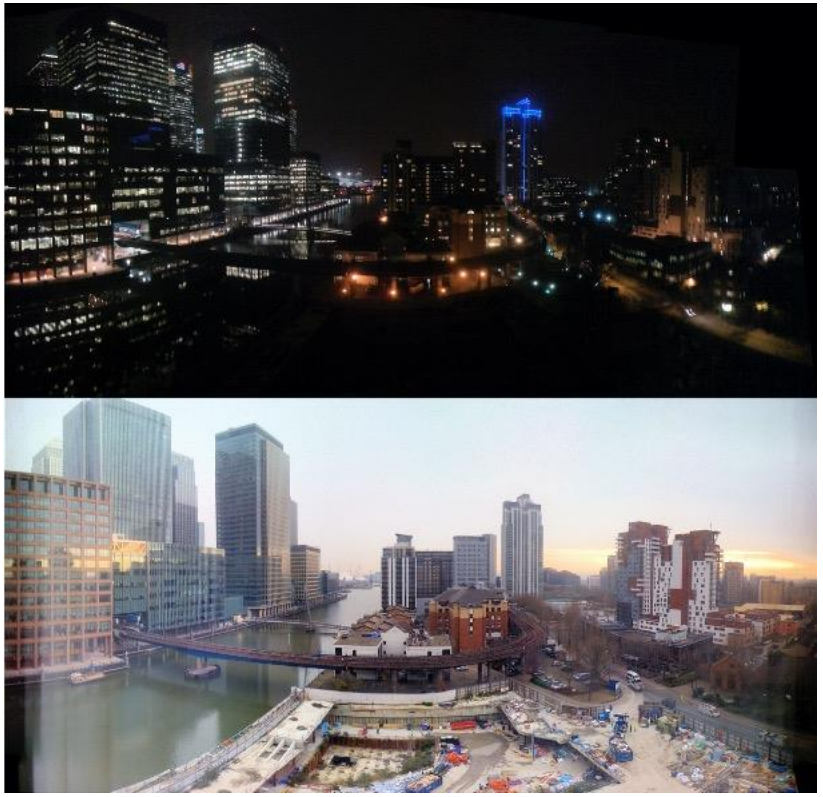
- Human color perception has both **strengths** and **limitations**, many of which are relevant to user-interface design
- For example:
  - Our vision is optimized to detect contrasts (edges), not absolute brightness
  - Our ability to distinguish colors depends on how colors are presented
  - Some people have color-blindness

The photoreceptors: rods and cones

- The retina has two types of **light receptor cells**

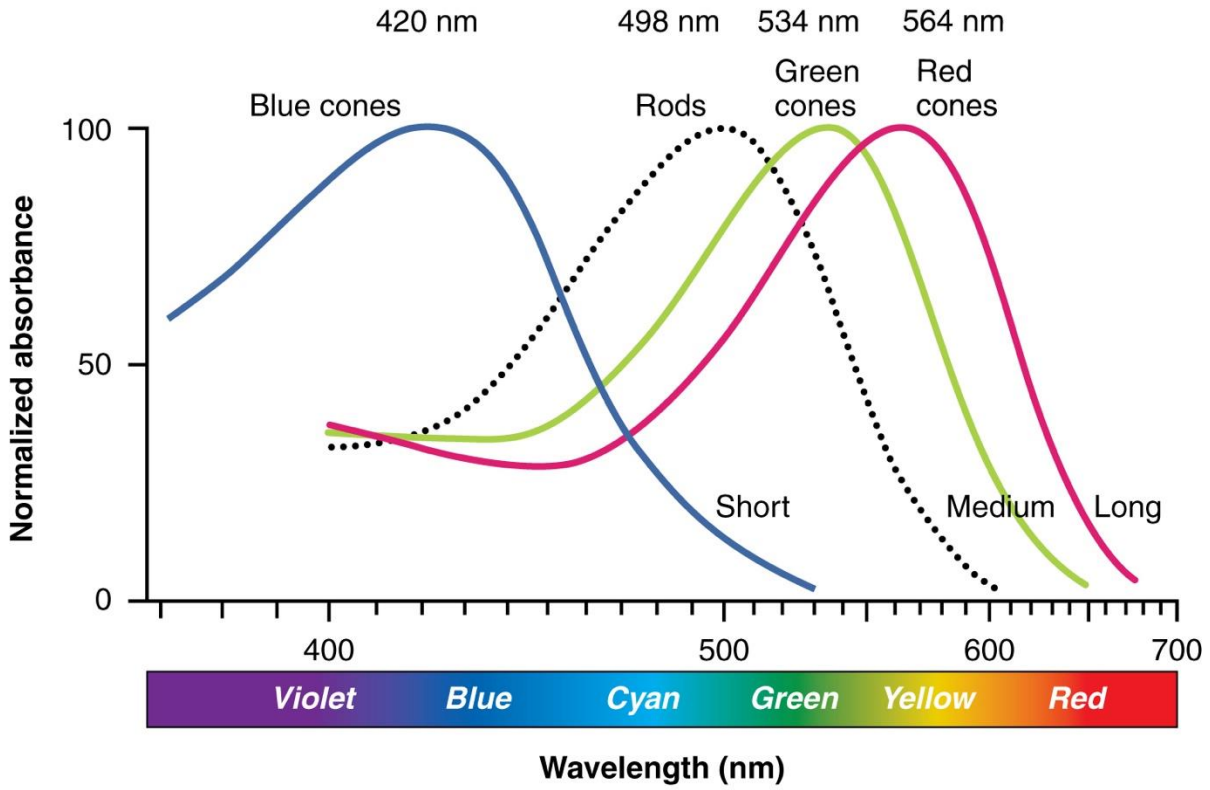
- **Rods:** are sensitive to the overall brightness, are responsible for vision at low light levels, and do not mediate color vision

- **Cones:** allow us to see fine details in bright light and give us the sensation of color



Color vision: the cones

- There are **3 types of cones**, which are sensitive to different frequencies/wavelengths of light
  - Low frequency (most sensitive to yellow and red)
  - Medium frequency (most sensitive to green)
  - High frequency (most sensitive to blue)



How color vision works

- The brain **combines** the signals from the cones, which enables us to see colors
- This combination results in "color-opponent channels"
  - Red-green difference signal channel
  - Yellow-blue difference signal channel
  - Black-white difference signal channel

Vision is optimized for contrast, not brightness

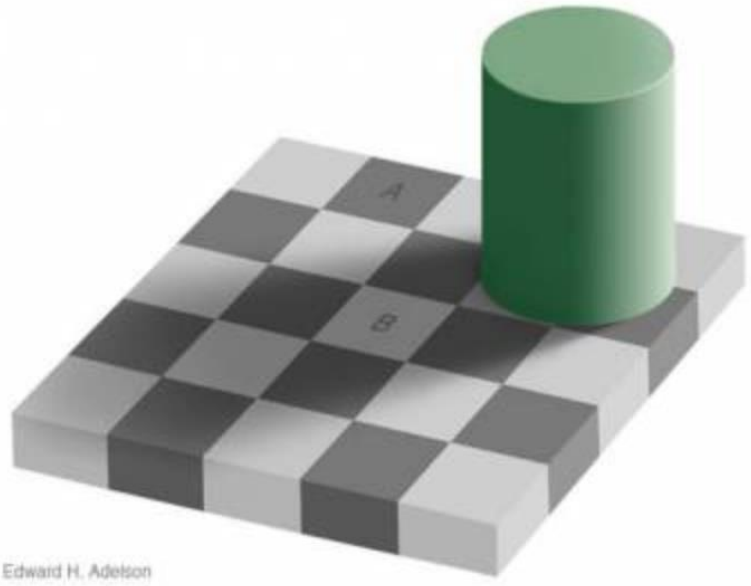
- Our visual system much more sensitive to differences in color and brightness — that is, to contrasting colors and edges — than to absolute brightness levels



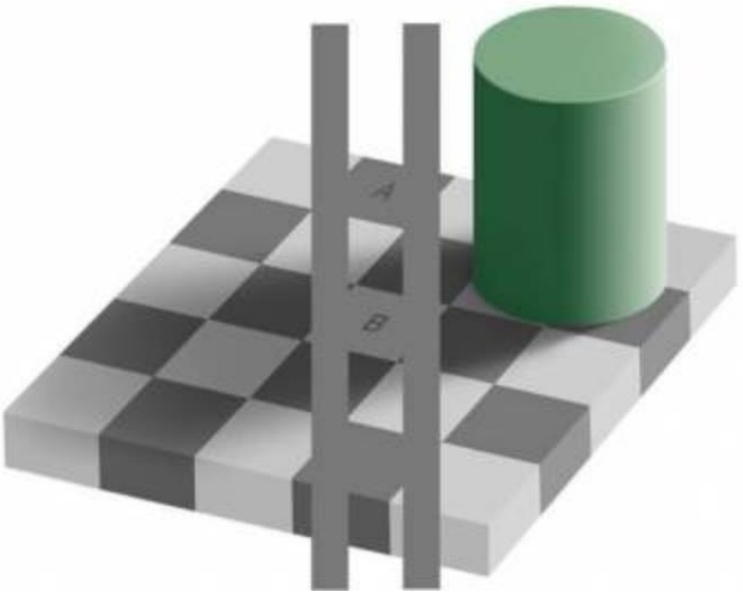
FIGURE 4.2 The inner gray bar looks darker on the right, but in fact is all one shade of gray.

- To our contrast-sensitive visual system, the bar looks lighter on the left and darker on the right because the outer rectangle is darker on the left and lighter on the right

Vision is optimized for contrast, not brightness: the Checker Shadow Illusion



The original image of the illusion
The squares marked A and B are the same shade of gray, yet they appear different.



The original image plus two stripes
By joining the squares marked A and B with two vertical stripes of the same shade of gray, it becomes apparent that both squares are the same.

- Video: https://youtu.be/z9Sen1HTu5o

The ability to discriminate colors depends on how colors are presented

- Three presentation factors affect our ability to distinguish colors from each other:
  - **Paleness.** The paler (less saturated) two colors are, the harder it is to tell them apart
  - **Color patch size.** The smaller or thinner objects are, the harder it is to distinguish their colors
  - **Separation.** The more separated color patches are, the more difficult it is to distinguish their colors, especially if the separation is great enough to require eye motion between patches

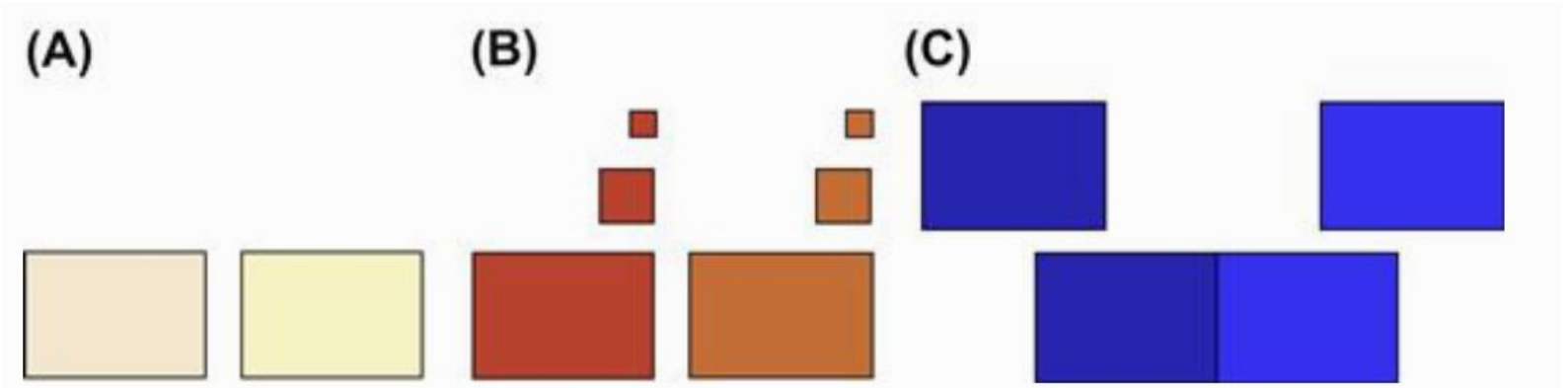


FIGURE 4.4 Factors affecting ability to distinguish colors: (A) paleness, (B) size, and (C) separation.

The ability to discriminate colors depends on how colors are presented

- Take into account the paleness in your design when choosing the colors



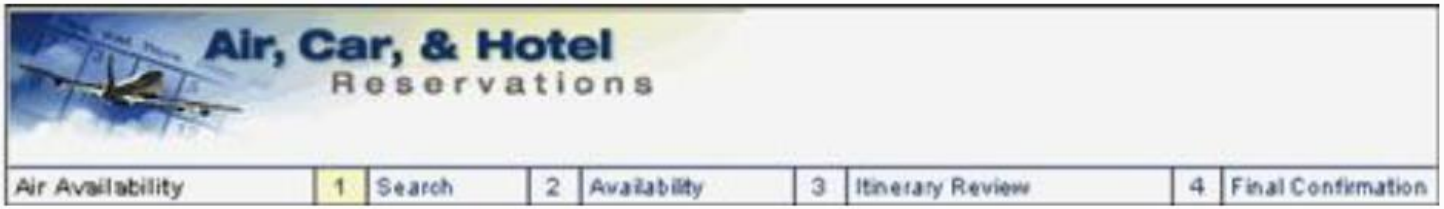


FIGURE 4.5 The pale color marking the current step makes it hard for users to see which step in the airline reservation process they are on in ITN.net's 2003 website.

- Small color patches are often seen in data charts and plots
- Many business graphics packages produce legends on charts and plots, but make the color patches in the legend very small
- Color patches in chart legends should be large to help people distinguish the colors

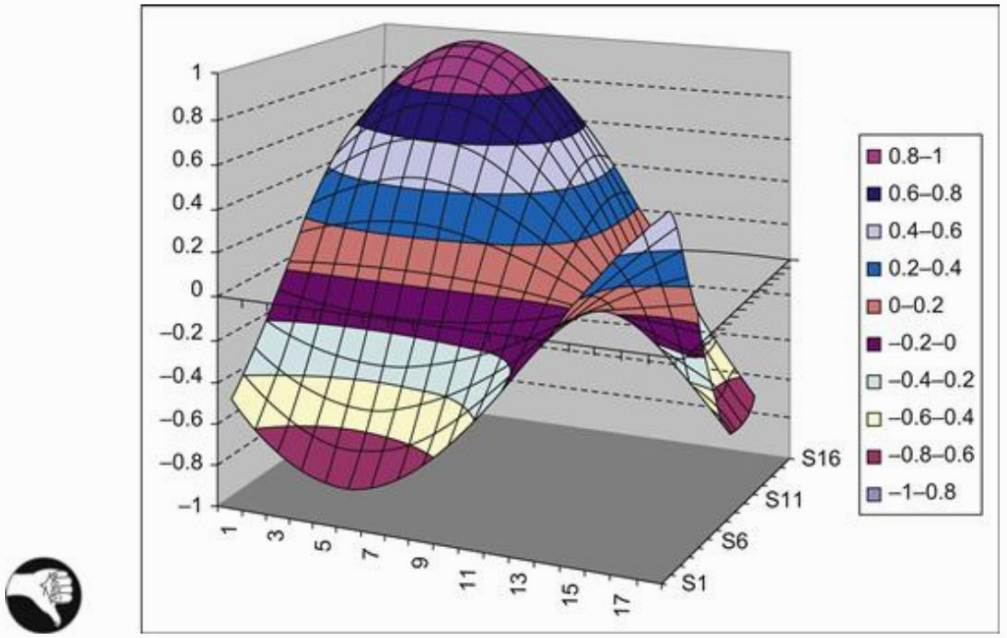


FIGURE 4.6 Tiny color patches in this chart legend are hard to distinguish.

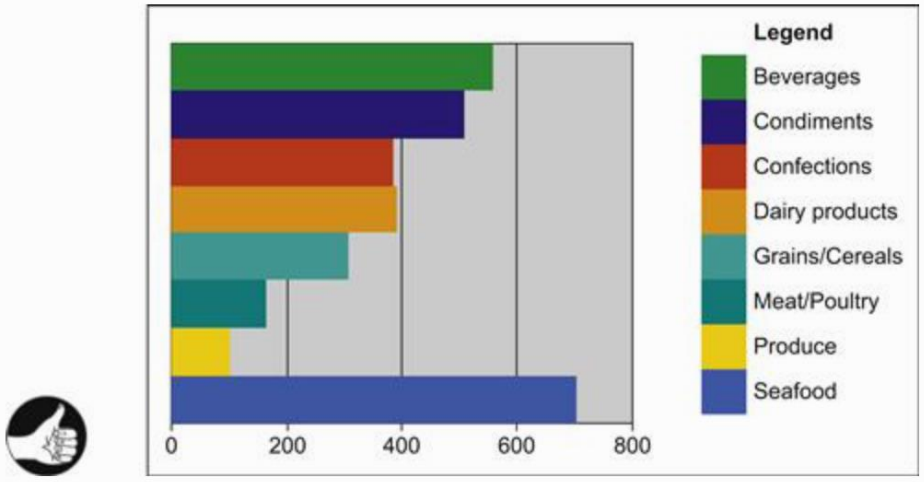


FIGURE 4.7 Large color patches make it easier to distinguish the colors.



FIGURE 4.8 The difference in color between visited and unvisited links is too subtle in MinneapolisFed.org's website.

- Text is often thin, so the exact color of text is often hard to determine
- Choose appropriate contrasting color for visited and unvisited links

Color-blindness

- Color blindness is not the inability to see colors, but a **deficit in the discrimination of certain pairs of colors** (8% of men and 0.5% of women)
- **Red-green** color blindness is the most common form (called protanopia, deuteranopia, protanomaly, and deuteranomaly)
- Blue-yellow color blindness less common

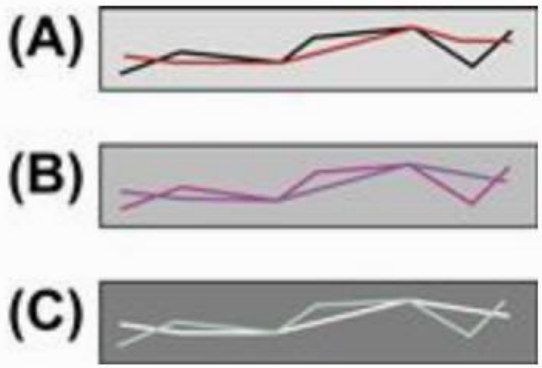


FIGURE 4.9 Red–green color-blind people can't distinguish (A) dark red from black, (B) blue from purple, and (C) light green from white.

Color-blindness



FIGURE 4.10 MoneyDance's graph uses colors some users can't distinguish.

- **Online color-blindness simulator**
  - https://www.color-blindness.com/coblis-color-blindness-simulator/

5 Guidelines for using color

1) **Distinguish colors by saturation and brightness, as well as hue**
2) **Use distinctive colors**
3) **Avoid color pairs that color-blind people cannot distinguish**
4) **Use color redundantly with other cues**
5) **Separate strong opponent colors**

1) Distinguish colors by saturation and brightness, as well as hue

- Avoid subtle color differences. **Make sure the contrast between colors is high** (but see guideline 5)
- One way to test whether colors are different enough is to view them in grayscale. If you can't distinguish the colors when they are rendered in grays, they aren't different enough

2) Use distinctive colors

- Our visual system combines the signals from retinal cone cells to produce three color-opponent channels: red–green, yellow–blue, and black–white (luminance)
- The colors that people can distinguish most easily are those that cause a strong signal (positive or negative) on one of the three color-perception channels, and neutral signals on the other two channels



FIGURE 4.13 The most distinctive colors: black, white, red, green, yellow, blue. Each color causes a strong signal on only one color-opponent channel.

- All other colors cause signals on more than one color channel, so our visual system cannot distinguish them from other colors as quickly and easily as it can distinguish those six colors

3) Avoid color pairs that color-blind people cannot distinguish

- Such pairs include
  - dark red versus black
  - dark red versus dark green
  - blue versus purple
  - light green versus white
- Don't use dark reds, blues, or violets against any dark colors. Instead, use dark reds, blues, and violets against light yellows and greens
- Use an **online color-blindness simulator** to check web pages and images to see how people with various color-vision deficiencies would see them

4) Use color redundantly with other cues

- **Don't rely on color alone**. If you use color to mark something, mark it another way as well

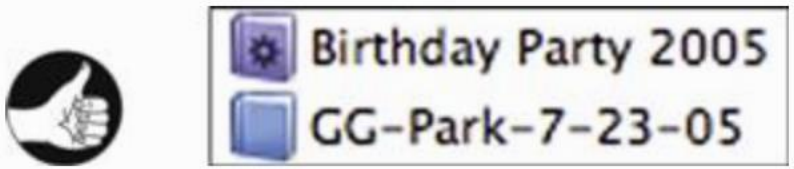


FIGURE 4.14 Apple's iPhoto uses color plus a symbol to distinguish two types of albums.

5) Separate strong opponent colors

- Placing opponent colors right next to or on top of each other causes a disturbing shimmering sensation, and so it should be avoided

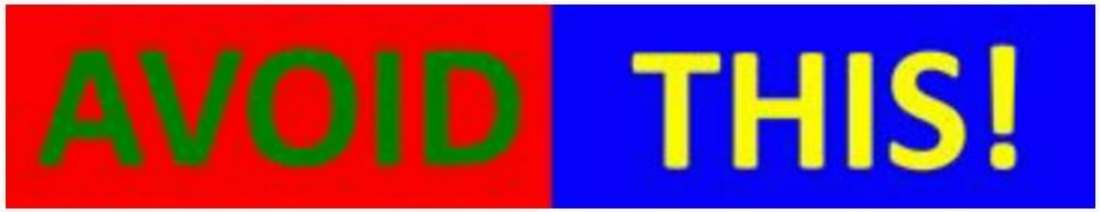


FIGURE 4.15 Opponent colors, placed on or directly next to each other, clash.

The color theory

- How do we choose the colors for our design?

- Color theory is a term used to describe the collection of rules and guidelines regarding the use of color **in art and design**

- Color theory informs the design of **color schemes**, aiming at aesthetic appeal and the effective communication of a design message **on both the *visual* level and the *psychological* level**

The color wheel

- The relationship between colors can be shown through use of the color wheel
  - It was first developed by Sir Isaac Newton in 1666



- Modern color theory is heavily based on Isaac Newton's color wheel, which displays three categories of colors:
  - **Primary colors**
  - **Secondary colors**
  - **Intermediate or tertiary colors**

- A thorough understanding of the color wheel and the relationship between colors enables designers to understand color better and know how to choose colors for their designs

The color wheel



- **Primary colors** (red, blue, yellow)
  - These are basic colors that cannot be broken down into any simpler colors
- **Secondary colors** (created by mixing two primary colors)
  - The secondary colors are orange, green, and purple
  - Mixing yellow and red creates orange; mixing blue and yellow creates green, and mixing blue and red creates purple
- **Intermediate or tertiary colors** (created by mixing primary and secondary colors)
  - The combination forms a hybrid, such as yellow-orange, red-orange, red-purple, blue-purple, blue-green, and yellow-green

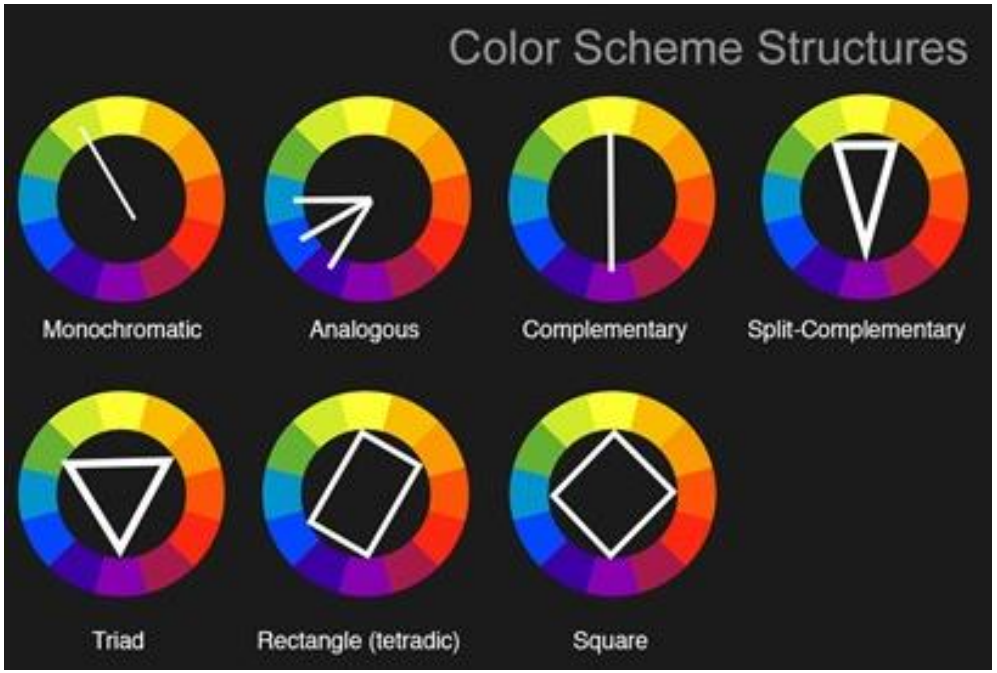
Achieving harmony in color

- Colors should be chosen to deliver an enhanced aesthetic appeal and a better user experience
- That means: it's a good idea to **think about what color scheme you will use at the start of the design process**

- How do we achieve harmony in our design using color?
- The way that colors are combined can either add to the look and feel or detract from it

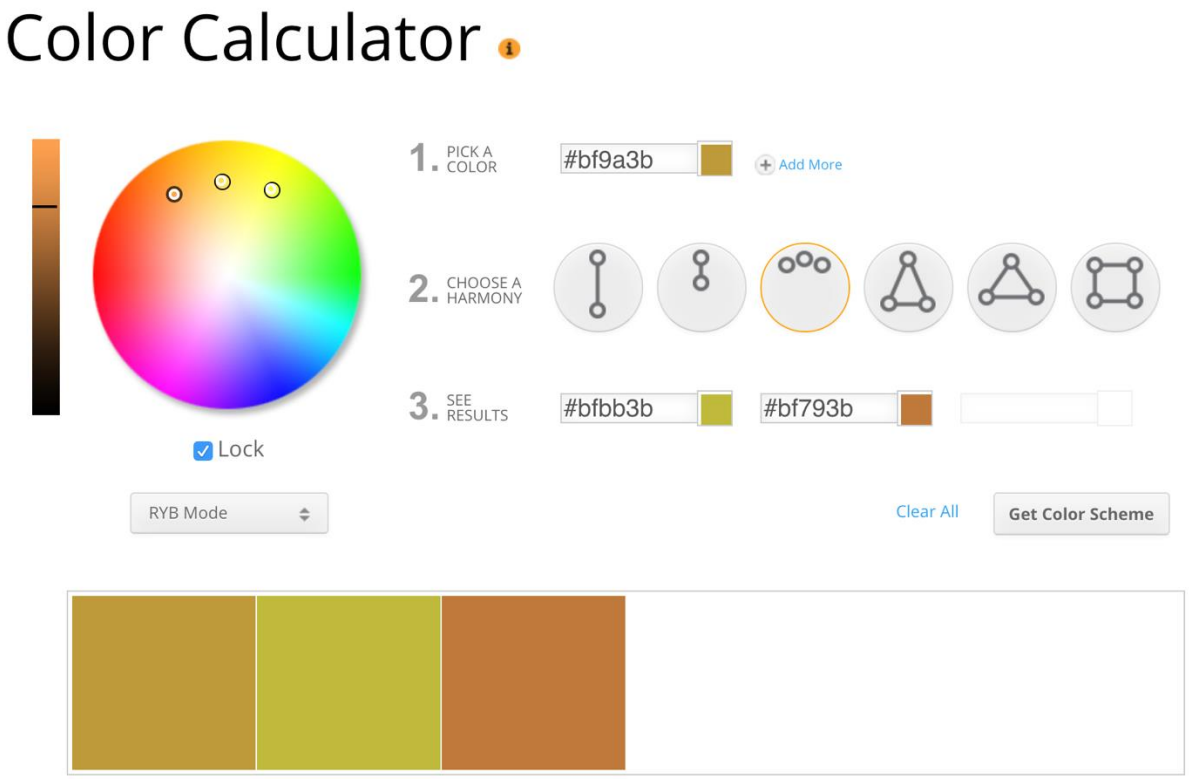
Achieving harmony in color: color schemes

- Colors can be combined to form **color schemes** that allow designers to achieve harmony in their designs
  - **Monochromatic**
  - **Analogous**
  - **Complementary**
  - **Split-complementary**
  - **Triadic**
  - **Tetradic**



Color wheel tools

- https://www.sessions.edu/color-calculator/
- https://www.rapidtables.com/web/color/color-wheel.html
- https://color.adobe.com/create



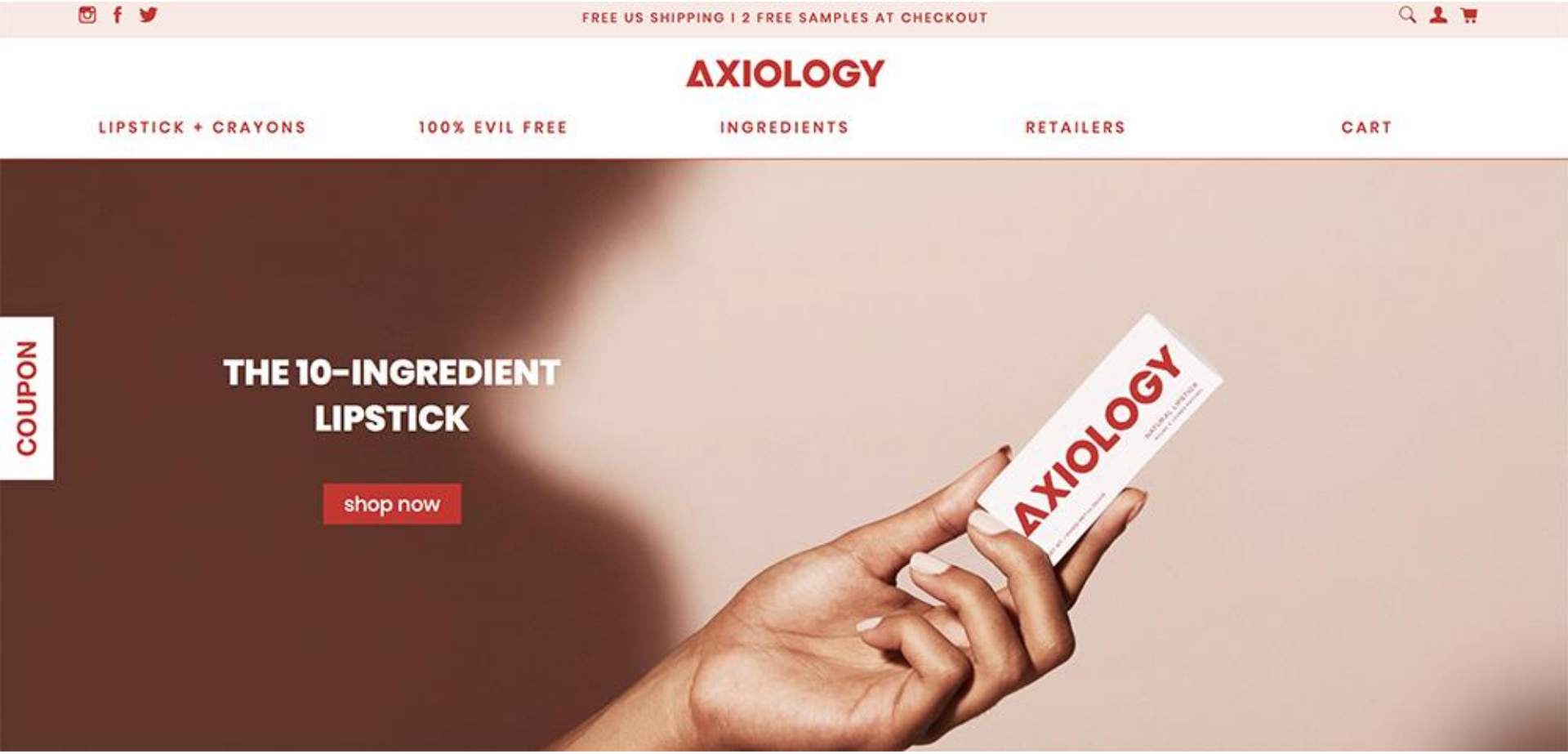
Monochromatic scheme



- It is based on the colors created from different tints (created by adding black or white to the original color), tones, and shades of one hue
- A monochromatic scheme is commonly used in minimal designs because one hue should result in a less distracting layout
- On the other hand, this scheme means that you cannot use multiple colors to help with visualizing information in the User Interface
  - That is the only price of simplicity

Monochromatic scheme

- This website uses a soft red as the main hue, with various lighter shades to complement



Analogous color scheme



- Is based on three colors located next to each other on the color wheel (e.g., red, red-orange, and red-violet)
  - This scheme can easily be found in nature – just think of trees in the autumn as the leaves change color
- There is a variant on this scheme, the "high-key" analogous color scheme
  - It's achieved by mixing your analogous shades with white
  - The effect achieved is one where the colors seem to "shimmer" and "blur" into each other

Analogous color scheme

- Example of an analogous color scheme with shades of purple, blue, and pink

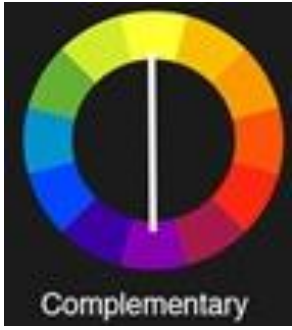


Analogous color scheme

- This website uses a bold blue and green analogous color palette that is pleasing to the eye



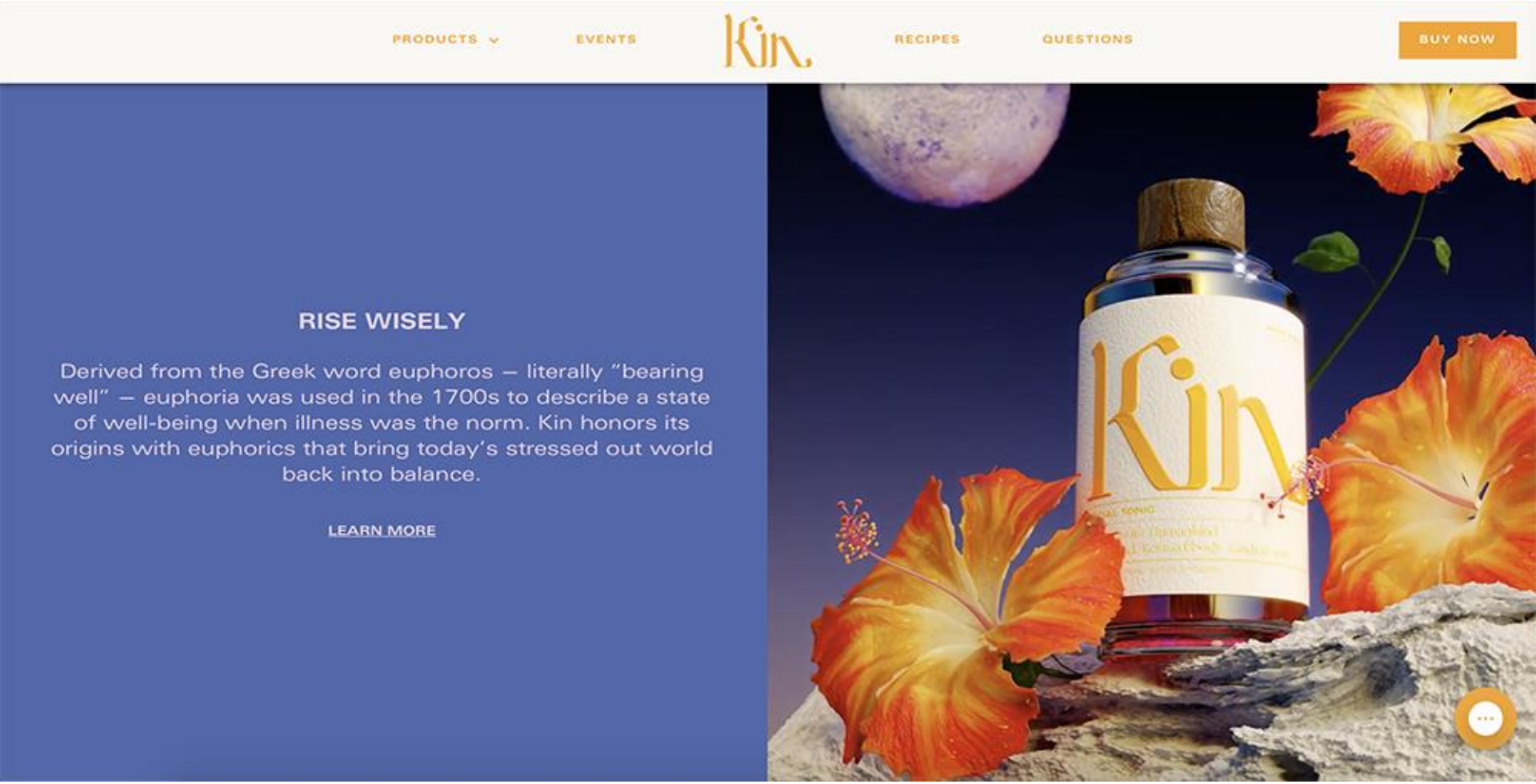
Complementary color scheme



- Uses one (or more) pairs of colors that, when combined, "cancel each other out"
  - E.g., when you combine the two colors, they produce white or black (or something very similar from the gray-scale)
- When you put two complementary colors next to each other, they show the greatest contrast
- In modern color theory, the pairs are
  - red/cyan
  - green/magenta
  - blue/yellow

Complementary color scheme

- This website uses its primary brand color orange as an accent, and pairs it with bold purple imagery for a dramatic effect



Split-complementary scheme



- This is a combination of using the complementary color scheme and the analogous color scheme
- In essence, complementary colors are chosen and then the colors on either side of them on the color wheel are also used in the design
- It's considered to soften the impact of a complementary color scheme, which can, in some situations, be too bold or too harsh on the viewer's eye

Split-complementary scheme



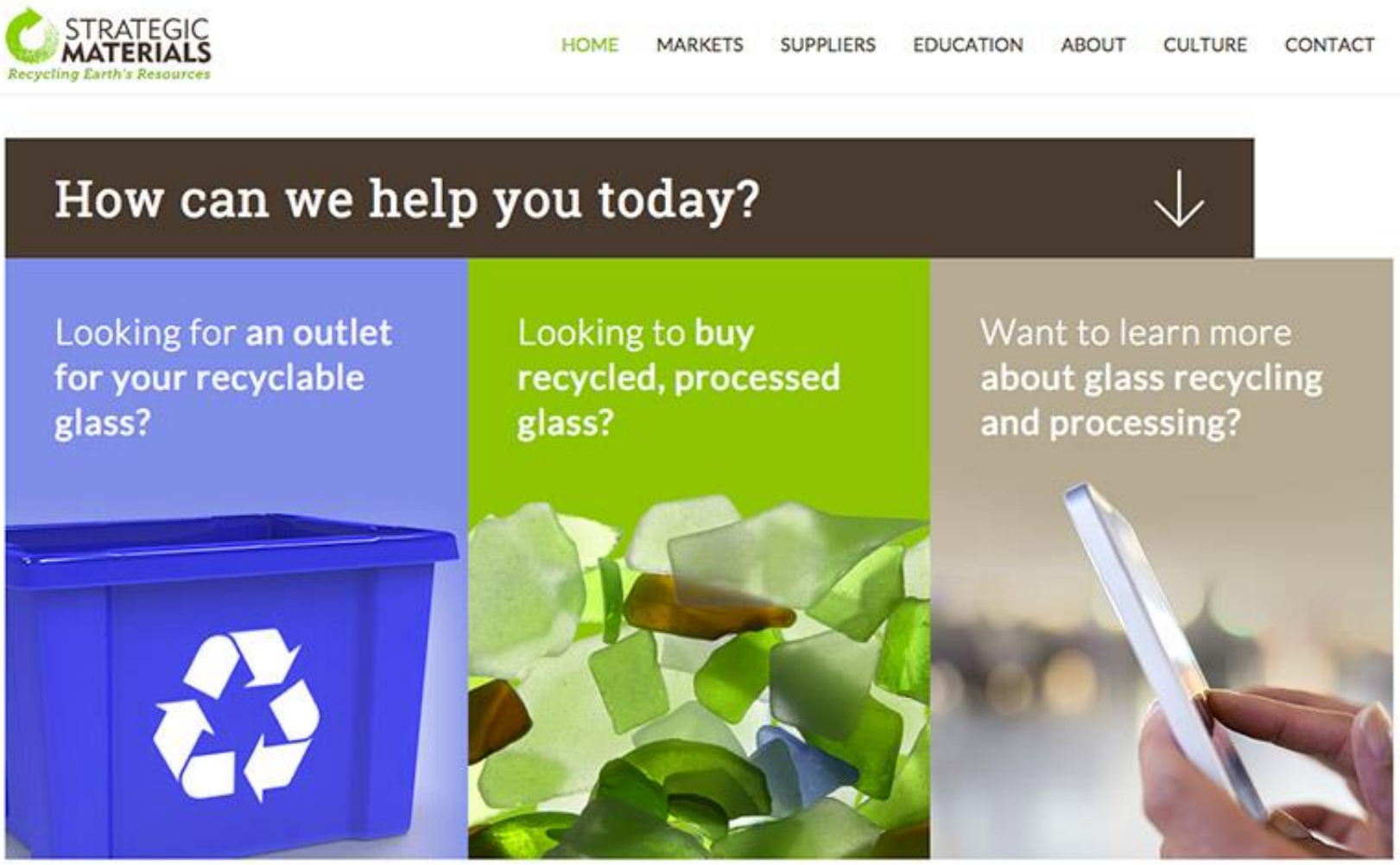
Triadic color scheme



- Is based on using three colors at equal distances from each other on the color wheel
- The easiest way to find a triadic scheme is to put an equilateral triangle on the wheel so that each corner touches one color
- These schemes are considered to be vibrant (even when the hues themselves are not) – they keep the harmony but deliver a high level of visual contrast
- You can find triadic schemes in a lot of art as it's easier to deliver a pleasing visual result with a triadic scheme than when using a complementary scheme

Triadic color scheme

- This website has a color palette based on a triadic color scheme

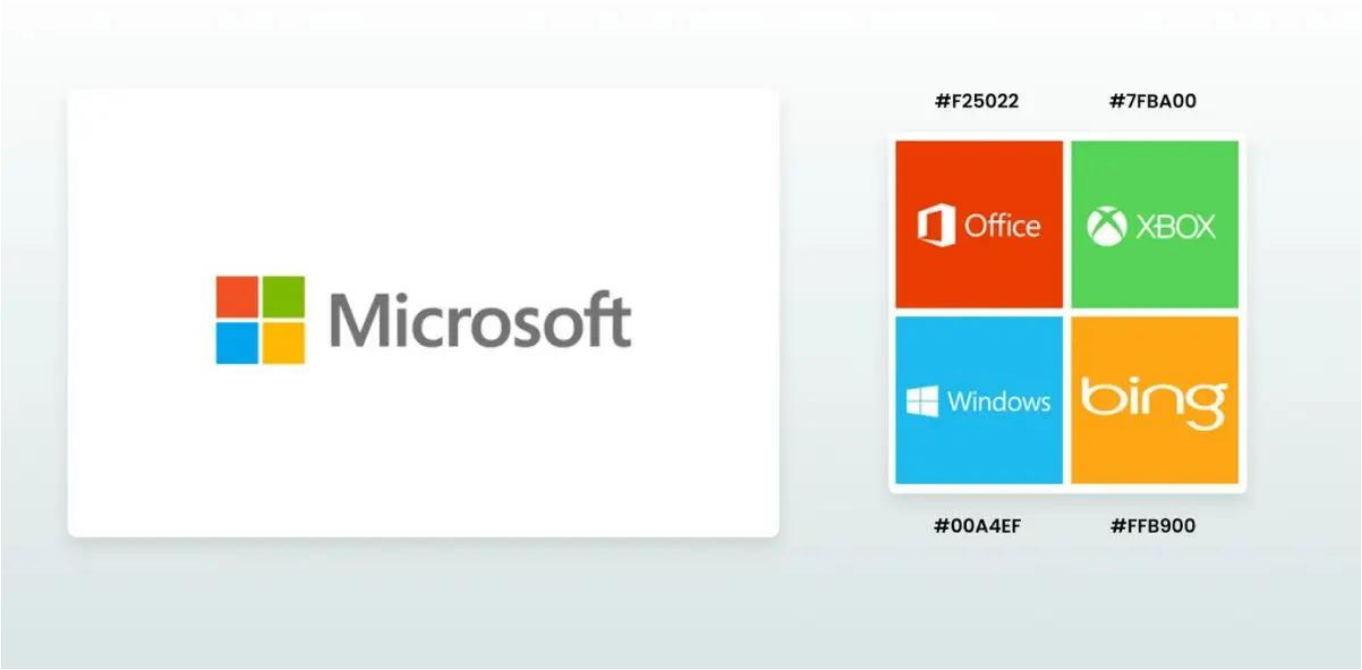


Tetradic color scheme



- Utilizes two sets of complementary pairs: four colors
- These can create very interesting visual experiences, but they are hard to keep in balance
  - Why?
  - It's because **one color of a tetradic scheme needs to dominate the other colors without completely overwhelming them**
  - An equal amount of each color often leads to a very awkward look, the last thing you want your users to see

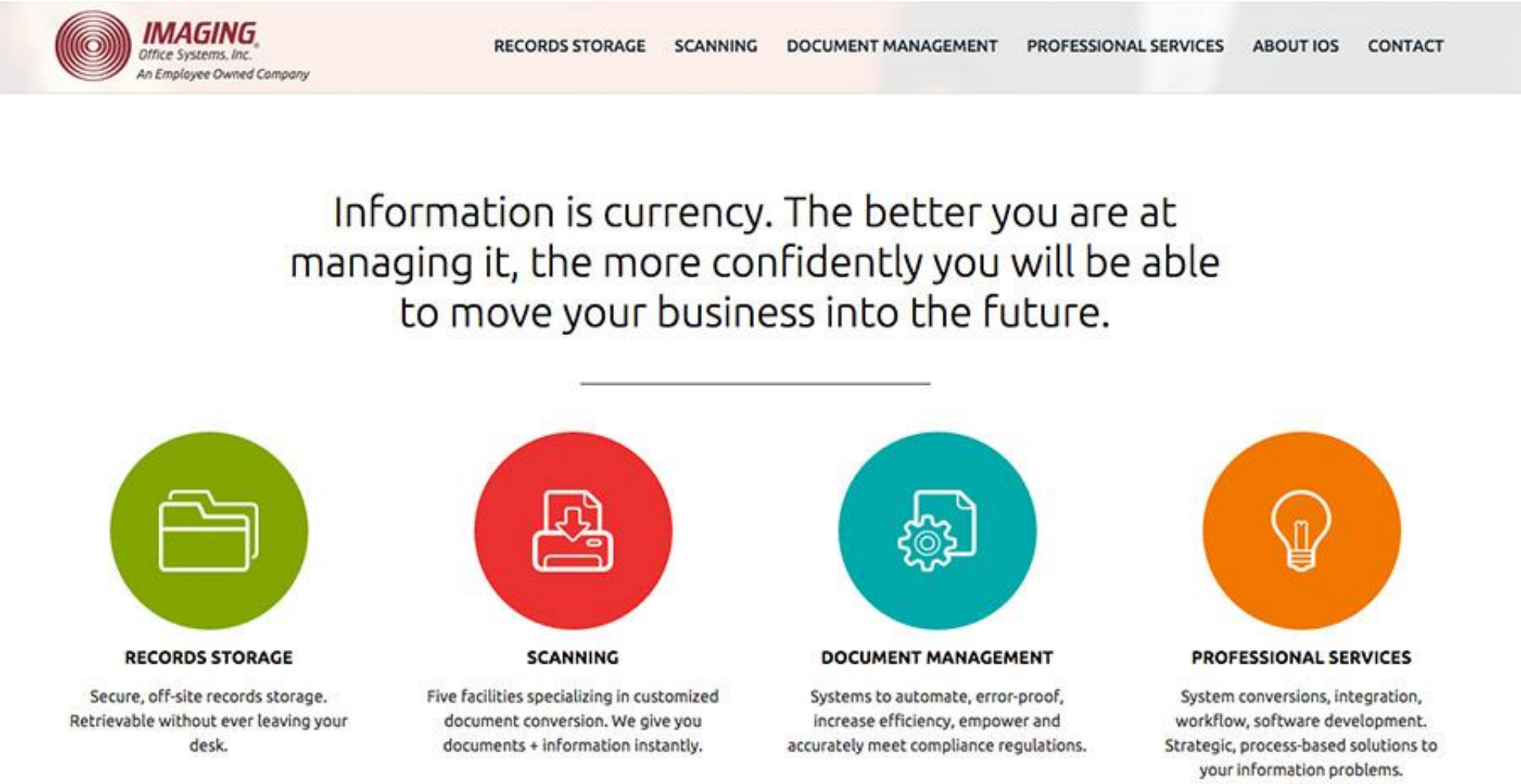
Tetradic color scheme



Google

Tetradic color scheme

- The color palette for this website is made up of two sets of complementary colors: red and green with blue and orange



Square color scheme



- Is a variant of the tetradic scheme
- Instead of choosing two complementary pairs, you place a square on the color wheel and choose the colors that lie on its corners
- Therefore, you'll find four colors that are evenly spaced at 90° from each other
- Unlike the tetradic color scheme, this approach often works best when all the colors are evenly used throughout the design

Square color scheme

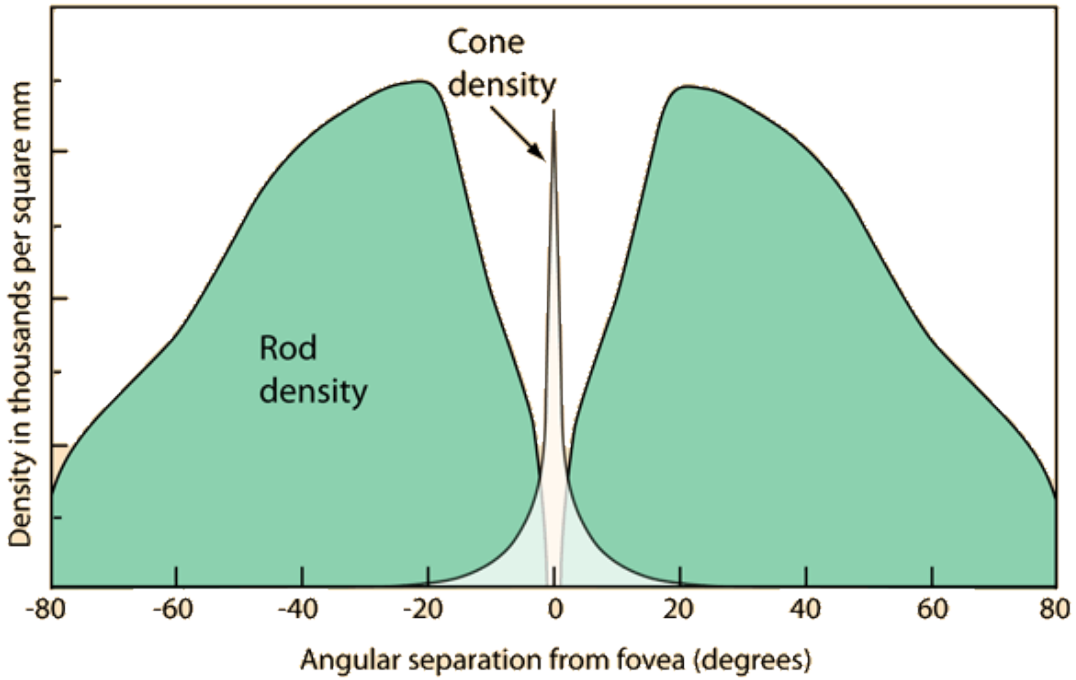




https://www.designmantic.com/blog/infographics/the-10-commandments-of-color-theory/

Our peripheral vision is poor

- Human vision has high resolution in a small area called "fovea" at the center of our visual field, and low resolution everywhere else
- Cones are concentrated in the fovea. Rods are absent there but dense elsewhere
  - We can discriminate colors better in the center of our visual field than at the edge



Our peripheral vision is poor

- Peripheral vision has a much lower resolution than foveal vision. It has three functions:
  1. **Guides the fovea**: peripheral vision provides low-resolution cues to guide our eye movements so that our fovea visits all the interesting and crucial parts of our visual field
  2. **Detects motion**: anything that moves in our visual periphery, even slightly, is likely to draw our attention – and hence our fovea – toward it
  3. **Lets us see better in the dark**: the cones do not function very well in low light condition

- Implications:
  - Stationary items in muted colors presented in the periphery of people's visual field often will not be noticed
  - Motion in the periphery is usually noticed

Error messages in peripheral vision

- The low acuity of our peripheral vision explains why software and website users fail to notice error messages in some applications and websites
- When someone clicks a button or a link, that is usually where his or her fovea is positioned
- Everything on the screen that is not within 1–2 centimeters of the click location (assuming normal computer viewing distance) is in peripheral vision, where resolution is low
- **If, after the click, an error message appears in the periphery, it should not be surprising that the person might not notice it**

Error messages in peripheral vision

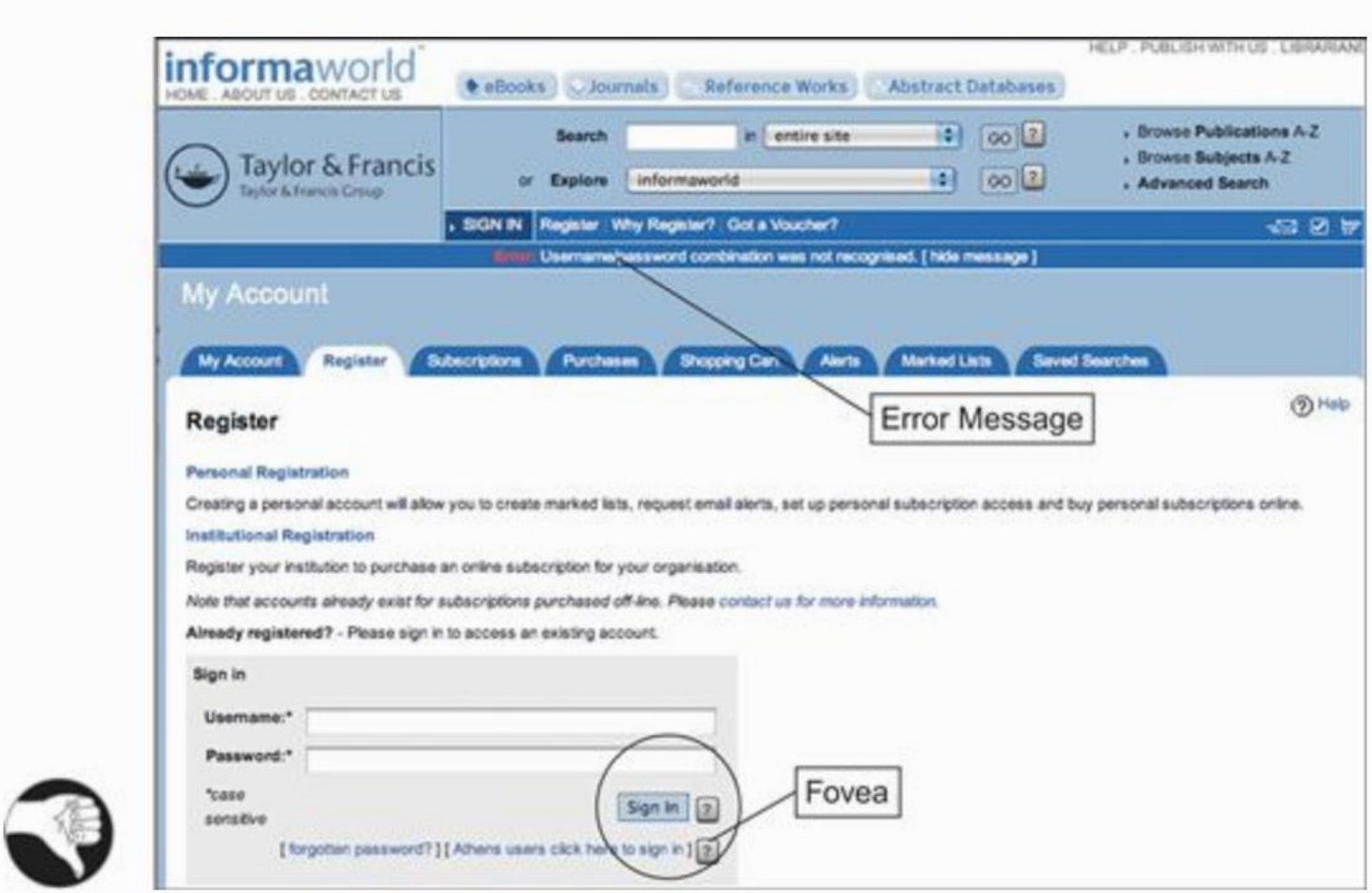


FIGURE 5.5 This error message for a faulty sign-in appears in peripheral vision, where it will probably be missed.

- This error message is entirely in red and near the "Login" button where the user's eyes are probably focused. Nonetheless, some users would not notice this error message when it first appeared



FIGURE 5.6 This error message for a faulty login is missed by some users even though it is not far from the "Login" button.

- Even though the error message is much closer to where users will be looking when they click the "Login" button, it is still in the periphery, not in the fovea
- The error message is not the only thing near the top of the page that is red. The page title is also red. Resolution in the periphery is low, so when the error message appears, the user's visual system may not register any change: there was something red up there before, and there still is

Common methods of making errors visible

- There are several common and well-known methods of ensuring that an error message will be seen
- **Put the error where users are looking**
  - People focus in predictable places when interacting with GUIs:
    - In Western societies, people tend to traverse forms and control panels from upper left to lower right
    - While moving the screen pointer, people usually look either at where it is or where they are moving it to
    - When people click a button or link, they can usually be assumed to be looking directly at it, at least for a few moments afterward
  - Designers can use this predictability to position error messages near where they expect users to be looking

Common methods of making errors visible

- **Mark the error**
  - Somehow mark the error prominently to indicate clearly that something is wrong
  - Often this can be done by simply placing the error message near what it refers to, unless that would place the message too far from where users are likely to be looking
- **Use an error symbol**
  - Make errors or error messages more visible by marking them with an error symbol, such as



Common methods of making errors visible

- **Reserve red for errors**
  - By convention, in interactive computer systems the color red connotes alert, danger, problem, error, etc.
  - Using red for any other information on a computer display invites misinterpretation
  - But suppose you are designing a website that has red as its main color: then you can use another color for errors, mark them with error symbols, or use stronger methods such as pop ups

Common methods of making errors visible



FIGURE 5.9 New member registration at AOL.com displays error messages prominently, near each error.

Stronger methods for making messages visible

- **Method 1: Pop-up message in error dialog box**
- **Method 2: use sound (e.g., beep)**
- **Method 3: wiggle or blink briefly**

- Note: use these 3 "heavy-artillery" methods sparingly to avoid **habituating** users
  - E.g., use them only for critical messages
  - Our brain pays less and less attention to any stimulus that occurs frequently

1) Pop-up message in error dialog box

- It interrupts the user's work and demands immediate attention: this, however, can annoy users

1. **Non modal pop-ups:** allow users to ignore them and continue working

2. **Application-modal pop-ups:** block any further work in the application that displayed the error, but allow users to interact with other software on their computer
   - should be used sparingly – e.g., only when application data may be lost if the user doesn't attend to the error

3. **System-modal pop-ups:** block any user action until the dialog has been dismissed
   - should be used extremely rarely – e.g., when the system is about to crash and take hours of work with it, or if people will die if the user misses the error message

1) Pop-up message in error dialog box

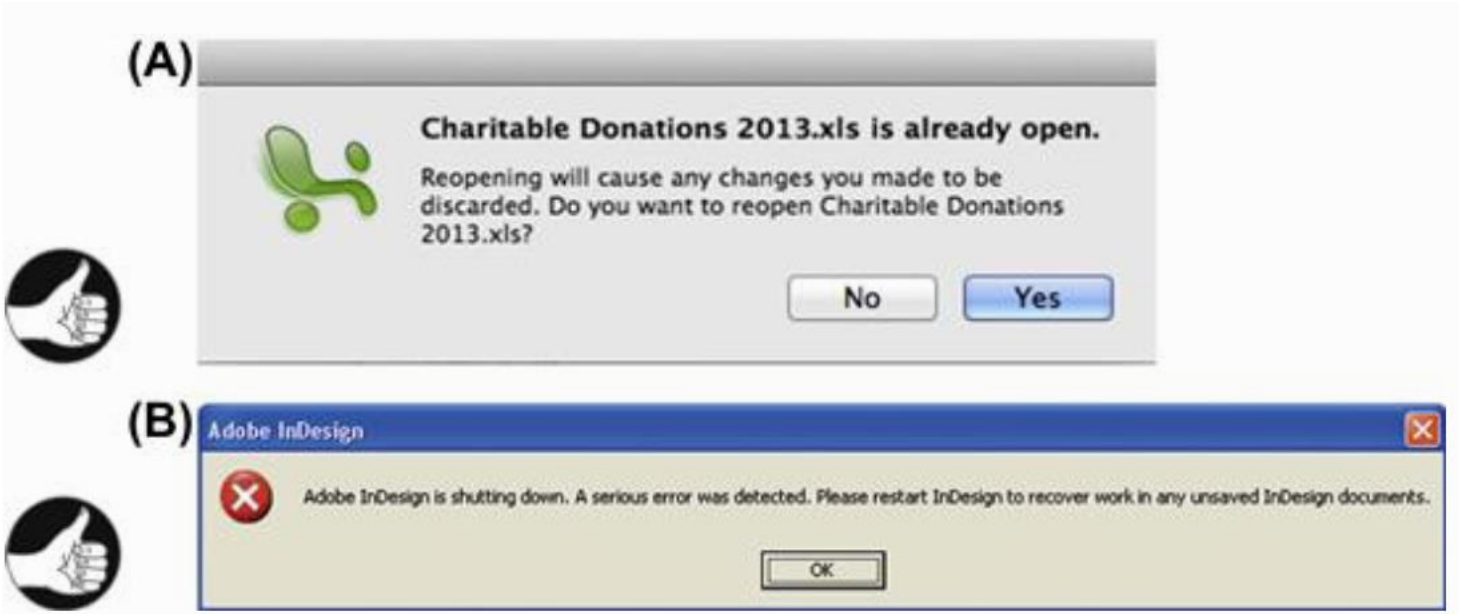


FIGURE 5.11 Appropriate pop-up error dialogs: (A) Microsoft Excel and (B) Adobe InDesign.

- Websites
  - Some people set their browsers to block all pop-up windows
  - If your website relies on pop-up error messages, some users may never see them

2) Use sound (e.g., beep)

- Beeps inform that something has happened that requires attention: the user's eyes reflexively begin scanning the screen for whatever caused the beep
- This can allow the user to notice an error message that is somewhere other than where the user was just looking, such as in a standard error message box on the display
- ….however,

2) Use sound (e.g., beep)

- ….however,
  - Beeps can be annoying in work environments where many users cause sounds for each error they make
  - In noisy work environments auditory signals emitted by an application might be masked by ambient noise
  - Some users mute the sound on their computers or turn it way down
- For these reasons, signaling errors and other conditions with sound are remedies that can be used only in very special, controlled situations
  - Computer games often use sound to signal events and conditions. In games sound is not an issue

3) Wiggle or blink briefly

- Our peripheral vision is good at detecting motion, and motion in the periphery causes reflexive eye movements that bring the motion into the fovea



- Designers can make use of this by wiggling or flashing messages briefly when they want to ensure that users see them
- It doesn't take much motion to trigger eye movement toward the motion. Just a tiny bit of motion is enough

FIGURE 5.12 Apple's iCloud shakes the dialog box briefly on login errors to attract a user's fovea toward it.

Reading

- HCI has studied extensively the reading process
- In this lesson we see some tips in order to facilitate the reading process of our users, useful for achieving a better UX

Poor information design can disrupt reading

- **Uncommon or unfamiliar vocabulary**
  - Words that the user doesn't know very well or at all
  - E.g., computer jargon
- **Difficult scripts and typefaces**

ABRAHAM LINCOLN'S GETTYSBURG ADDRESS

FOUR SCORE AND SEVEN YEARS AGO OUR FATHERS BROUGHT FORTH ON THIS CONTINENT, A NEW NATION, CONCEIVED IN LIBERTY, AND DEDICATED TO THE PROPOSITION THAT ALL MEN ARE CREATED EQUAL.

NOW WE ARE ENGAGED IN A GREAT CIVIL WAR, TESTING WHETHER THAT NATION, OR ANY NATION SO CONCEIVED AND SO DEDICATED, CAN LONG ENDURE. WE ARE MET ON A GREAT BATTLE-FIELD OF THAT WAR. WE HAVE COME TO DEDICATE A PORTION OF THAT FIELD, AS A FINAL RESTING PLACE FOR THOSE WHO HERE GAVE THEIR LIVES THAT THAT NATION MIGHT LIVE. IT IS ALTOGETHER FITTING AND PROPER THAT WE SHOULD DO THIS.

BUT, IN A LARGER SENSE, WE CAN NOT DEDICATE -- WE CAN NOT CONSECRATE -- WE CAN NOT HALLOW -- THIS GROUND. THE BRAVE MEN, LIVING AND DEAD, WHO STRUGGLED HERE, HAVE CONSECRATED IT, FAR ABOVE OUR POOR POWER TO ADD OR DETRACT. THE WORLD WILL LITTLE NOTE, NOR LONG REMEMBER WHAT WE SAY HERE, BUT IT CAN NEVER FORGET WHAT THEY DID HERE. IT IS FOR US THE LIVING, RATHER, TO BE DEDICATED HERE TO THE UNFINISHED WORK WHICH THEY WHO FOUGHT HERE HAVE THUS FAR SO NOBLY ADVANCED. IT IS RATHER FOR US TO BE HERE DEDICATED TO THE GREAT TASK REMAINING BEFORE US -- THAT FROM THESE HONORED DEAD WE TAKE INCREASED DEVOTION TO THAT CAUSE FOR WHICH THEY GAVE THE LAST FULL MEASURE OF DEVOTION -- THAT WE HERE HIGHLY RESOLVE THAT THESE DEAD SHALL NOT HAVE DIED IN VAIN -- THAT THIS NATION, UNDER GOD, SHALL HAVE A NEW BIRTH OF FREEDOM -- AND THAT GOVERNMENT OF THE PEOPLE, BY THE PEOPLE, FOR THE PEOPLE, SHALL NOT PERISH FROM THE EARTH.



FIGURE 6.7 Text in ALL CAPS is harder to read because we are not practiced at doing it. Outline typefaces complicate feature recognition. This example demonstrates both.

Poor information design can disrupt reading

- **Tiny fonts:** they make text hard to read
- **Text on noisy background**

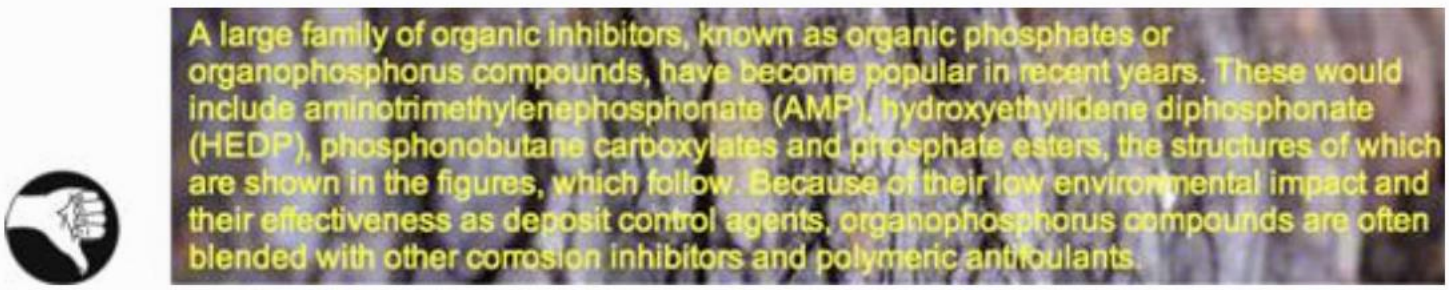


FIGURE 6.9 Arvanitakis.com uses text on a noisy background and poor color contrast.

Poor information design can disrupt reading

- **Information buried in repetition**

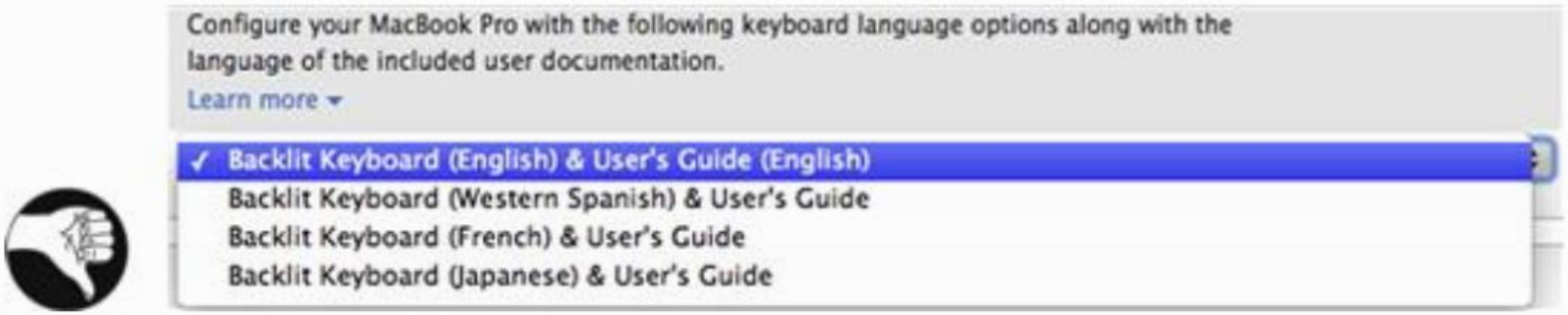


FIGURE 6.12 Apple.com's "Buy Computer" page lists options in which the important information (keyboard language compatibility) is buried in repetition.

Poor information design can disrupt reading

- **Centered or right-aligned text**
  - In fast reading our eyes are trained to go back to the same horizontal position and down one line



Exclusive Buyer Agency Offer

(No Cost) Service to Home Buyers!
Dan and Lida *want to work for you* ***if:***

.....

Would you like to avoid sellers agents who are pushing, selling, and trying to make sales quotas?
Do you want your agent to **be on your side** and **not** the sellers side?
Do you expect your agent to be responsible **and professional....?**
If you don't like to have **your time wasted,** Dan and Lida *want to work for you....*
If you understand *that everything we say and do, is to save you time, money, and keep you out of trouble....*
-and if you understand that some agents income and allegiances are in direct competition with your best interests....
-and if you understand that we take risks, give you 24/7 access, and put aside other paying business for you...
-and if you understand that we have a vested interest in helping you learn to make all the right choices...

- then, call us now, because Dan and Lida *want to work for you!!*

FIGURE 6.13 FargoHomes.com centers text, thwarting automatic eye movement patterns.

  - Each line of text starts in a different horizontal position
  - Automatic eye movements take our eyes back to the wrong place

Much of the reading required by software is unnecessary

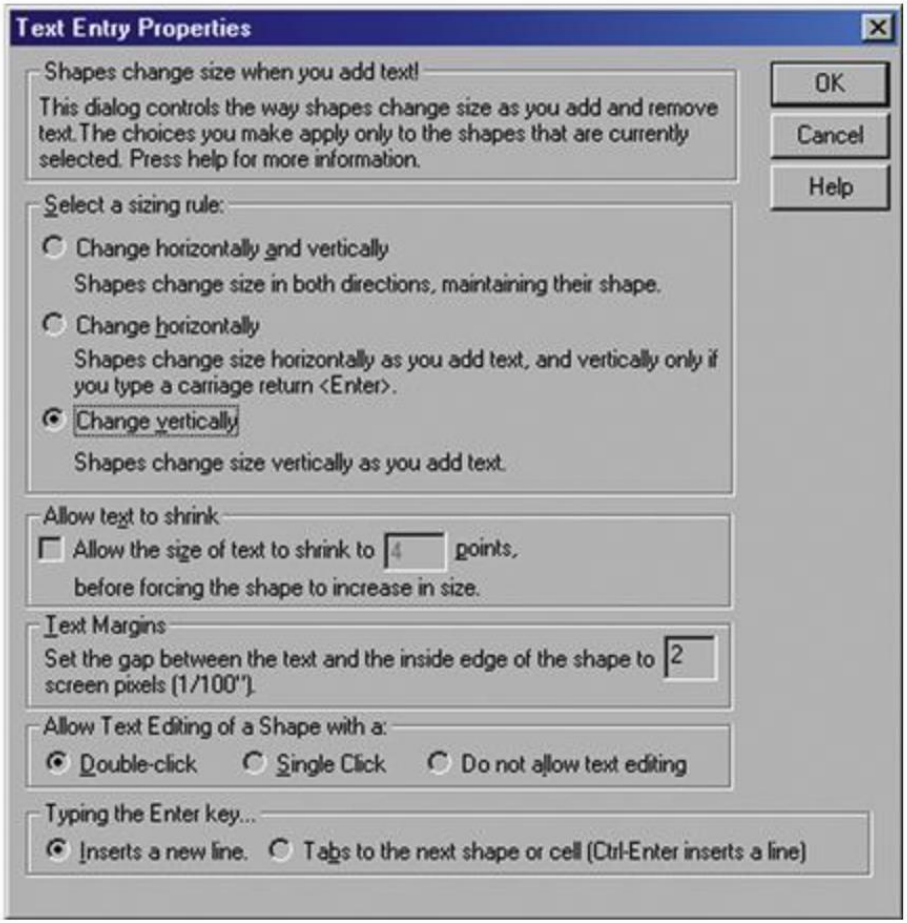
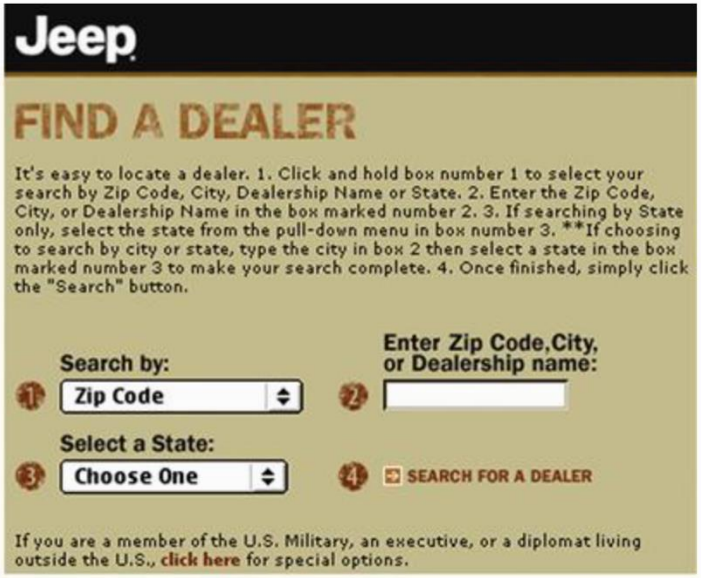


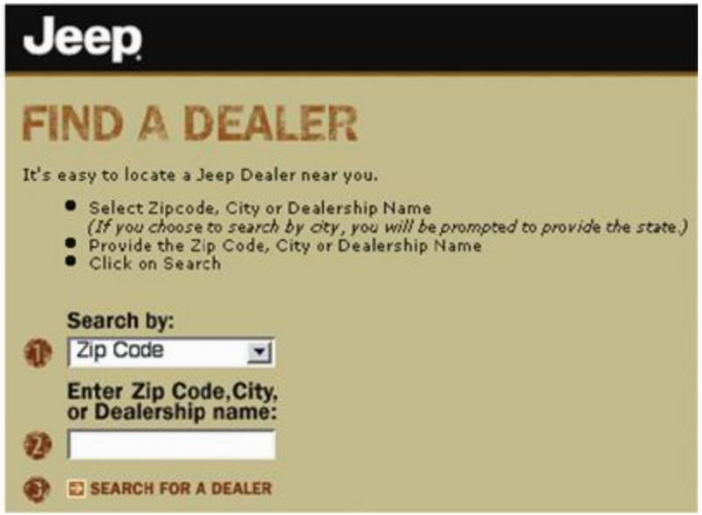
FIGURE 6.16 SmartDraw's "Text Entry Properties" dialog box displays too much text for its simple functionality.

- Instructions can often be **shortened** with no loss of clarity

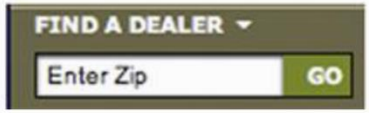
Much of the reading required by software is unnecessary



2002



2003



2007



2007



2009

FIGURE 6.18 Between 2007 and 2009, Costco.com drastically reduced the text in product descriptions.

Guidelines for reading

- Avoid disruptive flaws such as difficult or tiny fonts, patterned backgrounds, centering
- Use restricted, highly consistent vocabularies – sometimes referred to as "plain language"
- Format text to create a visual hierarchy to facilitate easy scanning: use headings, bulleted lists, tables, and visually emphasized words
- Minimize the need for reading
  - Don't present users with long blocks of prose text to read
  - In instructions, use the least amount of text that gets most users to their intended goals
  - In a product description, provide a brief overview of the product and let users request more detail if they want it

Lesson for the course "Human-Computer Interaction"

Laboratory:

Google Material Design (Part 1)

Prof. Luca Turchet



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OF TRENTO - Italy

Design Patterns

- A design pattern is a solution to a problem in a context
  - i.e., **it describes a problem, a solution, and where this solution has been found to work**
- Design patterns capture design experience
- One of the intentions of the patterns community is to create a **vocabulary** based on the names of the patterns, which designers can use to communicate with one another and with users
- A key characteristic of design patterns is that they can be instantiated or implemented in many different ways

Pattern language

- A pattern language is a network of patterns that reference one another and work together to create a complete structure
- Patterns are attractive to designers because they are tried and tested solutions to common problems
- It is common (although not obligatory) for pattern collections **to be associated with software components** that can be used with little modification, and as they are common solutions, many users are already familiar with them
- An example of interaction design guidelines and pattern libraries is Google Material Design

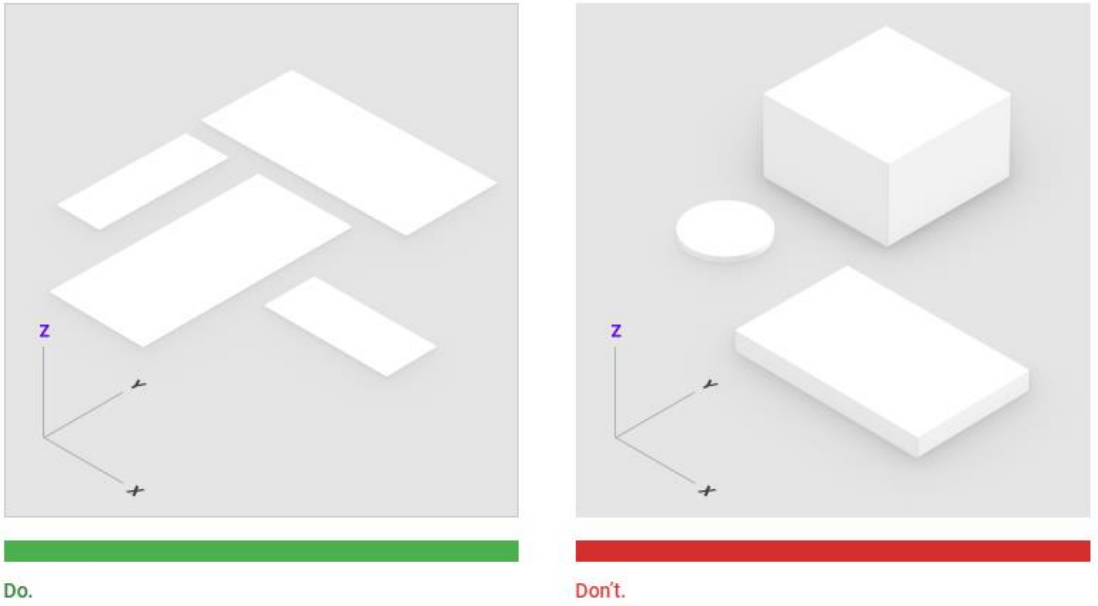
Google Material Design

- https://material.io/design/
- Is a visual design language that Google developed in 2014 (and updates it constantly)
- It synthesizes the classic **principles of good design** with the innovation of technology and science
- Check the goals and principles here: https://material.io/design/introduction/

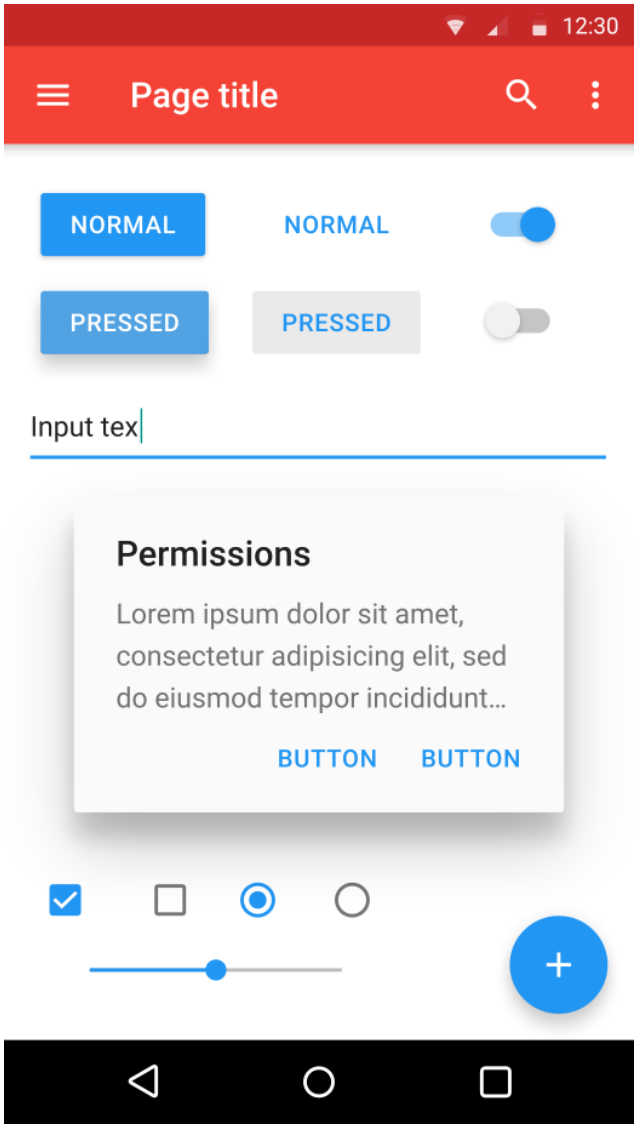
- Video: https://youtu.be/BUzUoc2ZOIo

Google Material Design

- Material Design is inspired by the physical world and its textures, including how they reflect light and cast shadows
- It uses a **metaphor** of 2D paper sheets that move on 3D



Google Material Design



- Example of a page of an Android app built with Material Design Guidelines, which shows buttons, toggle switches, a notification, check and radio buttons, a slider and a floating action button

- As of 2019, most of Google's mobile applications for Android had applied the new design language, including Gmail, YouTube, Google Drive, Google Docs, Sheets and Slides, Google Maps

Google Material Design

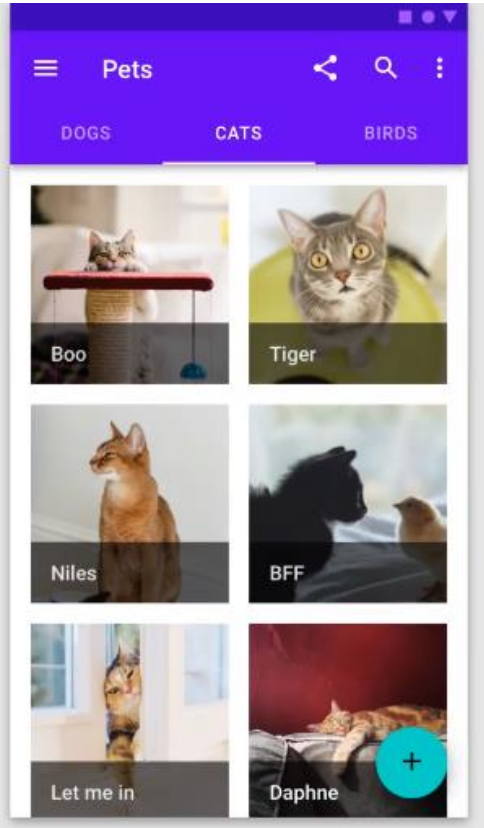
- We see it because it is useful to learn how to use a *style guide*: we will see the **Google Material Design Guidelines**
- It is associated to open source software libraries that simplify the development

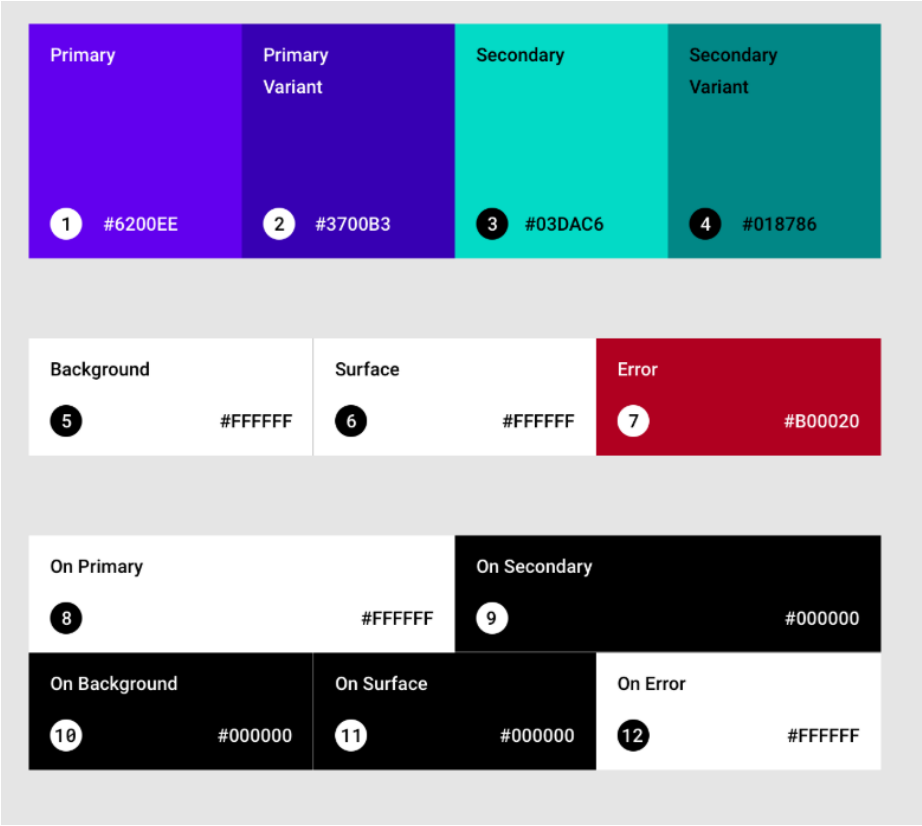
- Check the Material Design **case studies** (useful for finding inspiration and as a reference of guidelines applied in concrete scenarios)
  - https://material.io/design/material-studies/about-our-material-studies.html

The Color System

- The Material Design color system uses an organized approach to applying color to your UI:
  - A **primary** and a **secondary** color are typically selected to represent your brand
  - **Dark** and **light variants** of each color can then be applied to your UI in different ways
  - **Surface color**: affect surfaces of components, such as cards, sheets, and menus
  - **Background**: appears behind scrollable content
  - **Error colors**: indicates errors in components, such as invalid text in a text field
- https://material.io/design/color/the-color-system.html

The Color System





- Primary: the dominant color of the interface (the dark and light variants are used for the contrasts)
- Secondary: use it with parsimony and coherence (e.g., to enhance some widgets)

The Color System

- Your app's primary and secondary colors, and their variants, help create a color theme that is harmonious, ensures accessible text, and distinguishes UI elements and surfaces from one another

- To select primary and secondary colors, and generate light and dark variants of each, use:
  - **The Material Design palette tool:** https://material.io/resources/color/#!/?view.left=0&view.right=0
  - 2014 Material Design palettes: https://material.io/design/color/the-color-system.html#tools-for-picking-colors

Homework

- Analyze a website and an app and assess whether they comply with the guidelines for using color
- Analyze a programs and assess how it deals with error messages
- Analyze images in a website and assess whether it is accessible for color-blind people
  - **Use the online simulator**: https://www.color-blindness.com/coblis-color-blindness-simulator/
- Analyze a website and an app and assess whether they comply with the reading guidelines

Homework

- Start conceiving which **color schema** you will use for your app/website/software GUI
  - Consider the 10 commandments of color theory
  - https://www.designmantic.com/blog/infographics/the-10-commandments-of-color-theory/

Homework

- **Read the pages:**
  - https://material.io/design/color/the-color-system.html
  - https://material.io/design/color/applying-color-to-ui.html
  - https://material.io/design/color/color-usage.html
  - https://material.io/design/color/text-legibility.html

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- https://www.invisionapp.com/inside-design/website-color-schemes/